

TYLER YAN

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Education

University of Toronto

Expected Graduation: April 2026

Bachelor of Applied Science in Computer Engineering | Minor in Artificial Intelligence

Toronto, ON

- Dean's List: 1st, 2nd, and 3rd year | GPA: 3.65/4.00
- Courses: Software Design and Communication, Applied Fundamentals of Deep Learning, Operating Systems, Algorithms and Data Structures, Intro. to Machine Learning, Intro. to Artificial Intelligence, Computer Architecture

Work Experience

ePIC Blockchain Technologies

May 2024 – Aug 2025

Software Engineer Intern

Toronto, ON

- Achieved a 100% reduction in firmware validation test setup errors and inconsistent results by developing Autotest, a Rust/Python/YAML reliable testing framework to automate quality assurance and standardize pre-release verification
- Reduced MCU board debug time by 60% by creating a hardware validation application in Rust with automated boot-time tests and detailed failure diagnostics, improving clarity and reducing false positives for manufacturing technicians
- Improved system startup stability and mining efficiency by 35% through researching optimal temperature thresholds to prevent initialization in suboptimal conditions that caused dead chips, reduced hashrates, and startup failures
- Built a full-stack miner performance tracker using TypeScript, HTML/CSS, and Python with a FastAPI backend and SQL database, enabling users to monitor, visualize, export, and compare real-time data via a user-friendly interface
- Redesigned the Webdash UI using TypeScript, HTML, and CSS based on customer feedback, significantly improving miner management efficiency and monitoring transparency, resulting in high user satisfaction
- Mentored an intern for 4 months, providing structured onboarding, pair programming sessions, and code review support

Personal Projects

DePhi 0DTE

Apr 2025 – Present

- Leading a team of 4 software engineers to build a decentralized 0DTE trading platform on the Aptos blockchain, enabling 100% on-chain order placement, execution, and settlement, and reducing trading fees to almost zero via smart contracts
- Designed an intuitive frontend using TypeScript/HTML/CSS with real-time blockchain price feeds, wallet authentication, and position management tools, improving accessibility and usability for non-technical users
- Enabled low-latency (<1s), high-frequency trading while ensuring transparency and security by deploying gas-optimized Move smart contracts

SafeStride | [Source Code](#) | [Project Page](#)

Dec 2024 – Jan 2025

- Improved mobility safety and accessibility for visually impaired users by designing a two-device wearable system that translated visual input into directional haptic feedback and audio alerts for obstacles
- Developed the real-time obstacle detection system using OpenCV and YOLOv8, achieving a <100ms image-to-alert response time by optimizing Flask server-client communication across two mobile devices

FlightGAN: Image-to-Video Bird Animation | [Source Code](#) | [Research Paper](#)

Apr 2023 – Sep 2023

- Led a 4-member team to develop a generative adversarial network (GAN) model in Python that generated 300+ frame flying bird animations from static images, achieving an 85% realism rating in peer evaluations and a FID score of 23.1
- Designed a GAN architecture inspired by the StyleGAN2-ADA using PyTorch, incorporating adaptive data augmentation and style mixing
- Designed and implemented an automated scalable data pipeline using Python, OpenCV, and YOLOv8 to process 18,000+ image frames, increasing training and testing dataset quality and relevance while minimizing unwanted image noise

Leadership

University of Toronto - New College Don (Residence Assistant)

Aug 2023 – Present

- Responsible for a community of 100+ students, ensuring their safety, well-being, and social development
- Increased engagement in academic and social initiatives by 25% YoY by providing one-on-one mentorship and creating monthly programming strategies focused on mental health awareness and social connection

Technical Skills & Interests

Languages: Rust, Python, C, C++, Java, SQL, HTML5, CSS, TypeScript, JavaScript, Assembly, Swift

Developer Tools: Git, Linux, Unix, Docker, ReactJS, PyTorch, Pandas, MongoDB, NumPy, Flask, Bootstrap, scikit-learn

Interests: Basketball, Songwriting, Guitar, Chess, Juggling, Classic Rock, Sketching